

Dr. Nancy Modi

✉ nancy94251@gmail.com

☎ (+91) 7000336231

📍 Goa

🌐 www.linkedin.com/in/nancy-modi/



EDUCATION

2020 - 2026	Ph.D. CGPA: 8.76	Communication and Signal Processing Department of Electrical Engineering Indian Institute of Technology Bombay (IIT Bombay), Mumbai, India
2017 - 2019	M.Tech. CGPA: 8.92	Digital Communication Department of Electronics and Communication Maulana Azad National Institute of Technology Bhopal, India
2013 - 2017	B.E. CGPA: 7.98	Department of Electronics and Communication ITM Group of Institutions, Gwalior, India

PATENTS

[1] **N. Modi**, B. M. Kumar and J. Mukherjee, "Systems and methods for reducing reflections of radar radio waves," Indian Patent No.: 521364, March 7, 2024. **(Granted)**

[2] **N. Modi**, B. M. Kumar and J. Mukherjee, "A Non-Uniform Matching Layer Design for Wide-Angle Matching Automotive Bumpers," Application No.: 202521054675 dt. 06.06.2025. **(Published)**

RESEARCH PUBLICATIONS

JOURNALS

[1] **N. Modi**, B. M. Kumar and J. Mukherjee, "Design of Wideband and Wide-Angle Matching Bumper to Improve Automotive Radar Performance," in *IEEE Sensors Letters*, vol. 8, no. 5, pp. 1-4, May 2024, Art no. 6005304, doi: 10.1109/LENS.2024.3392431.

[2] B. M. Kumar, N. Kalva, **N. Modi** and J. Mukherjee, "Eight-Element Self-Isolated MIMO Antenna Based on Wideband SIW Rectangular Ring Slot Antenna," *AEU - International Journal of Electronics and Communications, Elsevier*, pp. 154555, 2023, doi: 10.1016/j.aeue.2023.154555.

[3] **N. Modi**, B. M. Kumar and J. Mukherjee, "A Non-Uniform Matching Layer Design for Wide-Angle Matching Bumper in Automotive Radar Applications," in *IEEE Antennas and Wireless Propagation Letters*. **(to be submitted)**.

[4] B. M. Kumar, **N. Modi** and J. Mukherjee, "A Wideband SIW Cavity-Backed Bident-Shaped Slot Antenna," in *IEEE Antennas and Wireless Propagation Letters*. **(to be submitted)**.

CONFERENCES

[1] U. Singh, L. Samdup, **N. Modi**, J. Mukherjee and A. Sarkar, "Detecting Hemoglobin Structural Variations Through Microwave Dielectric Analysis Using Horn Antennas," *2025 IEEE Microwaves, Antennas, and Propagation Conference (MAPCON)*, Kochi, India, 2025, pp. 1-4, doi: 10.1109/MAPCON65020.2025.11426425.

[2] **N. Modi**, S. Verma, B. M. Kumar and J. Mukherjee, "Investigation of Bumper Thickness to Study Automotive Radar Performance for Wide Incidence Angles," *2025 URSI Asia-Pacific Radio Science Meeting (AP-RASC)*, Sydney, Australia, 2025, pp. 1-4, doi: 10.46620/URSIAPRASC25/RPYZ6669.

[3] **N. Modi**, B. M. Kumar and J. Mukherjee, "Wideband Matching Automotive Bumper Using Frequency Selective Surface," *2024 IEEE Asia-Pacific Microwave Conference (APMC)*, Bali, Indonesia, 2024, pp. 169-171, doi: 10.1109/APMC60911.2024.10867544.

[4] B. M. Kumar, **N. Modi** and J. Mukherjee, "Circularly Polarized MIMO Antenna Based on SIW Cavity-Backed Square Ring Slot," *2024 IEEE Asia-Pacific Microwave Conference (APMC)*, Bali, Indonesia, 2024, pp. 480-482, doi: 10.1109/APMC60911.2024.10867831.

- [5] **N. Modi**, B. M. Kumar and J. Mukherjee, "A 45-Degree Rotated Square Ring Frequency Selective Surface Structure for Broadband Matching Automotive Bumper," *2024 URSI Regional Conference on Radio Science (URSI-RCRS)*, Bhimtal, India, 2024, pp. 1-4, doi: 10.23919/URSI-RCRS63970.2024.11201117.
- [6] **N. Modi**, B. M. Kumar and J. Mukherjee, "Analysis of the Effects of Rainwater Covered Bumper on the Automotive Radar Signals," *2024 18th European Conference on Antennas and Propagation (EuCAP)*, Glasgow, United Kingdom, 2024, pp. 1-5, doi: 10.23919/EuCAP60739.2024.10501490.
- [7] B. M. Kumar, **N. Modi** and J. Mukherjee, "Self-Isolated MIMO Antenna Using SIW Cavity Antenna for Dual-Band (28, 38 GHz) Applications," *2024 18th European Conference on Antennas and Propagation (EuCAP)*, Glasgow, United Kingdom, 2024, pp. 1-5, doi: 10.23919/EuCAP60739.2024.10501330.
- [8] **N. Modi**, B. M. Kumar and J. Mukherjee, "Study of the Tilted Mounting Angle of Automotive Radar Antenna Installed Behind a Painted Bumper," *2023 IEEE Microwaves, Antennas, and Propagation Conference (MAPCON)*, Ahmedabad, India, 2023, pp. 1-5, doi: 10.1109/MAPCON58678.2023.10464176.
- [9] **N. Modi**, B. M. Kumar and J. Mukherjee, "A Five-Element MIMO Antenna Based on Wideband Triple-Resonance SIW Antenna for Vehicular Communications," *2023 IEEE Microwaves, Antennas, and Propagation Conference (MAPCON)*, Ahmedabad, India, 2023, pp. 1-5, doi: 10.1109/MAPCON58678.2023.10463901.
- [10] **N. Modi**, B. M. Kumar, J. C. Dash and J. Mukherjee, "Study of the Bumper Thickness and its Placement for Automotive Radar Applications," *2022 IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting (AP-S/URSI)*, Denver, CO, USA, 2022, pp. 772-773, doi: 10.1109/AP-S/USNC-URSI47032.2022.9887075.
- [11] **N. Modi** and J. Mukherjee, "Analysis of models to reduce complexity and simulation time of Multilayered Media for Automotive Radar Applications," *2021 IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting (APS/URSI)*, Singapore, 2021, pp. 2002-2003, doi: 10.1109/APS/URSI47566.2021.9704777.
- [12] J. C. Dash, Mihir Parikh, **N. Modi**, J. Mukherjee, Sajitha VR, T. SaiDeepak, and V. Dhoot, "Performance Evaluation of Automotive Radar in The Presence of Bumper with Multiple Paint Layers Using Bidirectional Loss Model," *2021 15th European Conference on Antennas and Propagation (EuCAP)*, Dusseldorf, Germany, 2021, pp. 1-5, doi: 10.23919/EuCAP51087.2021.9411167.
- [13] **N. Modi**, V. Chaurasia, A. Pandey, R. Malpani and N. Shrirao, "Diagnosis of Cancerous Nuclei on Digital HIs: A Review," *Proceedings of the International Conference on Advances in Electronics, Electrical & Computational Intelligence (ICAEEC) 2019*, doi: 10.2139/ssrn.3572916.
- [14] R. Malpani, V. Chaurasia, A. Pandey, **N. Modi** and N. Shrirao, "MRI Segmentation for CAD of Brain Tumor: A Review," *Proceedings of the International Conference on Advances in Electronics, Electrical & Computational Intelligence (ICAEEC) 2019*, doi: 10.2139/ssrn.3576459.
- [15] N. Shrirao, A. Pandey, V. Chaurasia, R. Gupta, **N. Modi** and R. Malpani, "A Review: Comparative Analysis of Chaotic Systems With Their Fractional Order Counterparts," *Proceedings of the International Conference on Advances in Electronics, Electrical & Computational Intelligence (ICAEEC) 2019*, doi: 10.2139/ssrn.3576454.

ACADEMIC PROJECTS

THESIS

Ph.D. Thesis Title: Performance Improvement of an Automotive Radar in the Presence of Bumper and Paint Layers (*Dissertation successfully defended on January 7, 2026*).

M.Tech. Thesis Title: Breast Cancer HI Image Segmentation

COLLABORATION WORK

Indian Institute of Technology Bombay–Mercedes-Benz Research and Development India (IITB-MBRDI) collaboration project on "Development of a Generic Mathematical Model for Matching Layer design between Radar and Bumper".
Years: 2020–2021

AREA OF INTEREST

Antenna Arrays, Automotive Radar, Radome, Substrate Integrated Waveguide Antennas, MIMO Antennas, Microstrip Antennas, 3D Printing, Circularly Polarized Antenna.

TEACHING INTEREST

Antenna Theory, Electromagnetics, Microwave Integrated and Circuits, Signals and Systems, Control Systems, Radar System, Network System.

SKILLS

SOFTWARE TOOLS: CST Studio Suite, MATLAB, Mathematica.

Operation of Vector Network Analyzer (VNA) and Spectrum Analyzer.

Basic operations of FDM 3D printer.

TEACHING ASSISTANT EXPERIENCE

Course	Course Instructor
EE 611 Microwave Integrated Circuits	Prof. Jayanta Mukherjee
EE 602 Radar Systems	Prof. Kushal R. Tuckley
EE 606 Fibre Optic Communication	Prof. R. K. Shevgaonkar
EE 799 Advanced Antennas	Prof. Jayanta Mukherjee
EE 338 Digital Signal Processing	Prof. Vikram M. Gadre
EE 785 Electromagnetic Interference & Electromagnetic Compatibility	Prof. Kushal R. Tuckley
Analog Circuits (NPTEL)	Prof. Jayanta Mukherjee
Microwave Integrated Circuits (NPTEL)	Prof. Jayanta Mukherjee

AWARDS

[1] Awarded certificate for '*Excellence in Teaching Assistantship*' from the Department of Electrical Engineering, IIT Bombay as a Teaching Assistant for 'Digital Signal Processing' course.

[2] Secured 93% in PDR and 97.3% in CDR in CANSAT Competition (2017) organized by the NASA, American Astronautical Society (AAS) and American Institute of Aeronautics and Astronautics (AIAA).

VOLUNTEER SERVICES

Reviewer	Scientific Reports, Springer Nature	Year: 2025
Reviewer	Space Aerospace and Defence Conference (SPACE)	Years: 2024 – 2026
Reviewer	Microwave, Antennas, and Propagation Conference (MAPCON)	Years: 2022 – 2025